Fig. 10: (L) Millets kept in a stainless-steel box with a small hole in the cover, (R) cooked millets

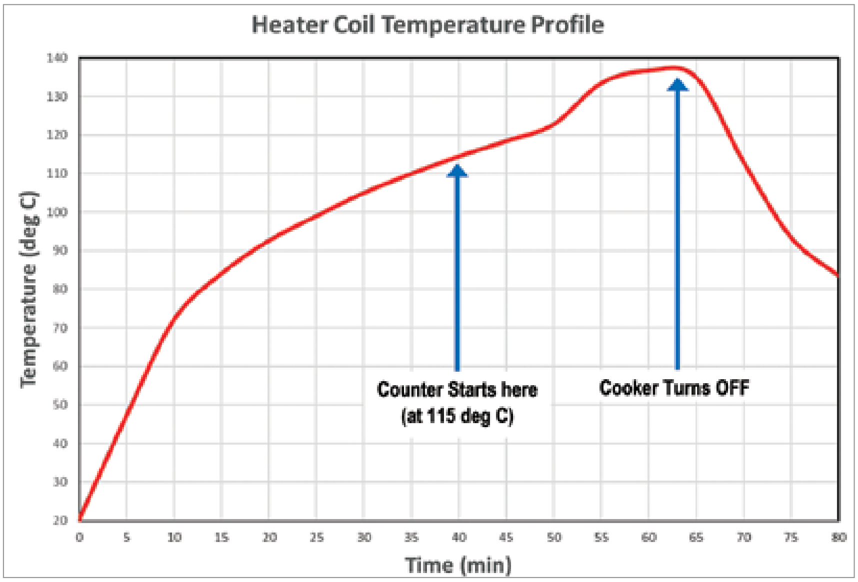

Fig. 11: Plot of heater coil temperature with time (elephant curve)

PARTS LIST

Semiconductors:	
IC1	- CD4060 binary counter with a built-in oscillator
IC2	- LM393 comparator
T1	- BC558 PNP transitor
T2, T3	- BC546 NPN transitor
D1, D4	- 1N4007 rectifier diode
D2, D3	- 1N4148 signal diode
LED1	- 5mm white LED
LED2	- 5mm green LED
LED3	- 5mm red LED
IRF1	- IRF540 MOSFET
Resistors (all 1/4-watt, ±5% carbon):	
R1, R3	- 22-kilo-ohm
R2	- 390-ohm, 1-watt
R4	- 4.7-mega-ohm
R5-R7	- 1.2-mega-ohm
R8-R10, R21, R24	- 4.7-kilo-ohm
R11	- 1-kilo-ohm
R12, R16, R17, R22, R23, R25	- 10-kilo-ohm
R13	- 47-kilo-ohm
R14	- 3.3-kilo-ohm
R15	- 100-ohm
R18, R19	- 470-ohm
R20, R26	- 1-mega-ohm
VR1	- 2.2-mega-ohm potmeter
VR2	- 1-kilo-ohm potmeter
Capacitors:	
C1	- 1000µF, 63V electrolytic
C2	- 47nF ceramic disk
C3	- 0.47µF ceramic disk
C4	- 4700µF, 16V electrolytic
C5	- 0.1µF ceramic disk
Miscellaneous:	
PZ1	- Buzzer
S1	- On/off switch
S2	- Push-to-on switch
CON1-CON4	- 3-pin connector
	- AC to DC converter 48V, 2A
	- DC to DC converter XL7015
	- Stainless-steel box
	- 15mm copper wire 5mm dia
	- Heater coil (1800-watt)
	- Three small wooden battens
	- Teflon wires
	- Heatsink for IRF540
	- 10mm thick poly urethane foam
	- Thermistor
	- Nut, bolt, wire etc
	- Casserole with stainless steel surface inside (200mm dia× 80mm depth)
	- Banana clips (4)
	- 3-leg heater support
	- Thermistor (10k NTC)
	- Three 18V, 30W solar panels (optional)
	- 150 grams of millets
	- 3-pin socket

Efficiency of AC-DC converter = 95% (assumed)

Total power consumption = 78.4/0.95 = 82.5W

Max cooking time = 1 hr 30 min (assumed)

Energy consumed = 82.5W × 1.5 hrs = 124Wh

Based on our calculations, it is evident that the proposed cooker is highly energy-efficient and does not consume power to maintain food warmth, potentially reducing energy costs. However, it has limitations, as it cannot cook tougher items like rice and dal. It is best suited for soaking-dependent food types like millets and certain vegetables like beans and carrots.

The prototype has a capacity of about 150 grams of millets, with stainless-steel box dimensions of 170mm in diameter and 50mm in height. Increasing the cooking capacity can be achieved by using a box with the same diameter but a height greater than 50mm. If the box touches the casserole cover, you can increase the PUF ring thickness to accommodate a taller stainless-steel box.

The proposed cooker requires a continuous power supply and is best suited for areas with infrequent power interruptions. In the event of a power outage during cooking, the user can resume by monitoring the steam from the copper pipe. When a steady stream of steam is visible, the cooking process is complete. Since it is a low-power device, it can also be connected to a home inverter for uninterrupted power.

On a sunny day, the cooker can